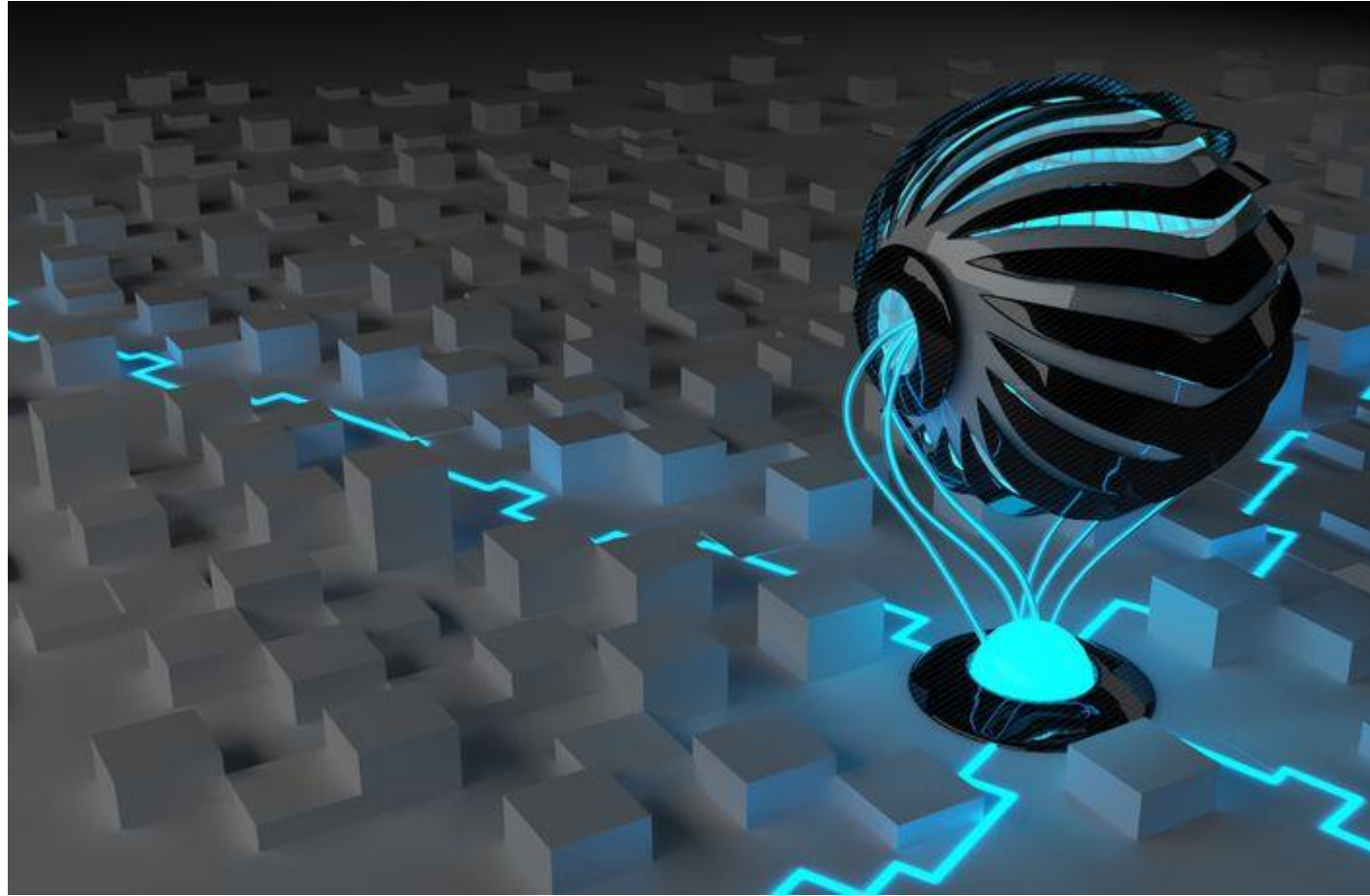


Computer Graphics and Visualization

Chapter 8

Latest Trend in Computer Graphics [6 hours]



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Content

- 8.1. Interactive Visualization
- 8.2. Distributed Scene Rendering
- 8.3. Augmented Reality(AR), Virtual Reality(VR) and Mixed Reality(MR)
- 8.4. Game Development and Real-Time Graphics
- 8.5. Applications of AR, VR and Gaming

1. Interactive visualization in computer graphics

- Interactive visualization in computer graphics refers to the use of visual representations of data or 3D models that allow users to interact with them in real time. This can include adjusting the view, modifying variables, or navigating within the visualized data. In computer graphics, this is typically achieved through techniques like 3D rendering, real-time simulation, and interactive user interfaces.

Here are some [common aspects of interactive visualization](#) in computer graphics:

1. 3D Visualization:

- **3D Models:** Users can rotate, zoom, and pan around 3D objects or environments (e.g., architectural models, virtual environments).
- **Animation:** Moving parts or elements within a scene can be dynamically altered based on user inputs.

2. User Interaction:

- **Mouse/Touch Input:** Allowing users to click, drag, or pinch to zoom in and out of the visualization.
- **Keyboard Shortcuts:** Providing hotkeys or other shortcuts for faster navigation or interaction.
- **Voice Commands:** Some systems use voice input to interact with visualizations, particularly in virtual reality (VR) or augmented reality (AR).

3. Real-Time Data Updates:

- **Dynamic Data Visualization:** The graphics can change as new data is provided, which is common in applications like scientific simulations, stock market visualizations, or weather models.

4. Virtual Reality (VR) & Augmented Reality (AR):

- VR can immerse the user in a 3D world, while AR overlays virtual objects onto the real world, allowing for more intuitive and hands-on interactions.

5. Graph and Network Visualizations:

- **Graph-based Structures:** Interactive exploration of networks, such as social media graphs, biological networks, or financial systems.
- **Zooming & Panning:** Users can zoom in on specific areas of a graph or move around large datasets to focus on specific regions.

6. Data Visualization:

- **Charts and Graphs:** Interactive charts where users can hover over or click on elements for more detailed information. For example, a time-series graph where users can zoom in on specific periods.
- **Heatmaps & Scatter Plots:** Visualizing large datasets where users can modify variables and immediately see how the data changes

Tools for Interactive Visualization:

- **WebGL & Three.js:** For 3D graphics and visualization in the browser, WebGL (JavaScript API for rendering 3D graphics) and Three.js (a 3D graphics library) are commonly used.
- **D3.js:** A powerful JavaScript library for creating interactive data visualizations (typically for 2D visualizations like bar charts, scatter plots, and pie charts).
- **Unity & Unreal Engine:** For building more complex 3D simulations, VR/AR environments, and interactive graphical interfaces.
- **Processing & P5.js:** Libraries that help create interactive graphics, often used for educational purposes or for rapid prototyping.
- **Blender:** Can be used for creating 3D visualizations and models that can be interacted with in various contexts.

Distributed rendering in computer graphics

- ❑ It refers to the process of spreading the computational load of rendering an image across multiple machines or devices in order to speed up the rendering process or to handle more complex scenes that would be too demanding for a single machine.
- ❑ It's especially useful for rendering photorealistic images or animations, which require heavy computations for simulations of lighting, textures, shadows, and other complex visual effects.

About Distributed Rendering

Distributed Systems

+

Computer Graphics
and Rendering

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Distributed Rendering

Performance Scaling

Data Scaling

Performance Scaling

- It is often advised to focus system design on hardware scalability rather than on capacity.
- It is typically cheaper to add a new node to a system in order to achieve improved performance than to partake in performance tuning to improve the capacity that each node can handle.
- Performance scaling therefore allows frames to be rendered more quickly

Data Scaling

- Data Scaling pertains to scaling and improving data handling rather than workload handling.
- This means that data scaling allows for increased sizes of datasets to be viewed.
- However, data scaling often falls prey to load balancing issues and latency.
- Despite this, different methods of distributing the workload tend to favor this type of scaling over the other.

"Distribution" in Distributed Rendering

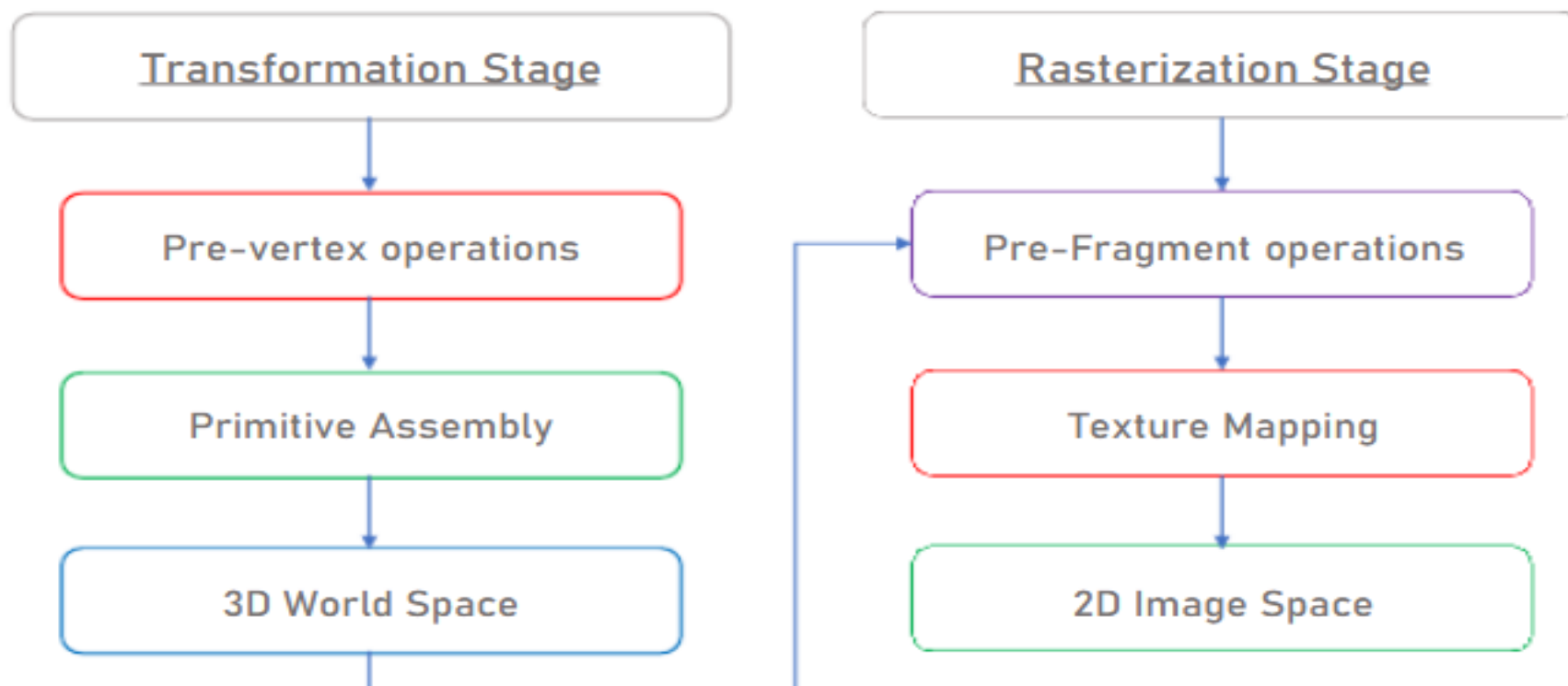
Frame Distribution

Pixel Distribution

Object Distribution

Hybrid Distribution

Distributed Rendering Pipeline



Advantages

Increasing rendering capability

Deal with larger datasets in smaller amounts of time

Combine multiple techniques in single visualization

Increase Interactivity

Disadvantages

Cannot break rendering pipeline as it is implemented in hardware

Usually suitable graphics driver and source code is unavailable

Additional Cost Overhead due to need of buffer

Virtual reality (VR),

- Virtual reality or virtual realities (VR), also known as immersive multimedia or computer-simulated reality, is a computer technology that replicates an environment, real or imagined, and simulates a user's physical presence and environment to allow for user interaction. Virtual realities artificially create sensory experience, which can include sight, touch, hearing, and smell.
- Most up-to-date virtual realities are displayed either on a computer monitor or with a virtual reality headset (also called head mounted display)
- Some simulations include additional sensory information and focus on real sound through speakers or headphones targeted towards VR users

Why Virtual Reality (VR)?

- VR is able to immerse you in a computer-generated world of your own making: a room, a city, the interior of human body. With VR, you can explore any uncharted territory of the human imagination.

Virtual Components

- Virtual reality (VR) systems are made up of hardware and software components that work together to create a virtual environment.

The **key components of a VR** system include:

1. Headset (HMD - Head-Mounted Display)

- The most recognizable component of VR, it displays the virtual world to the user.

Key Features:

- Displays the visual content through lenses.
- Tracks the movement of your head and adjusts the view accordingly.
- Includes built-in sensors, such as accelerometers, gyroscopes, and magnetometers, to track head orientation and position

2. Tracking System

- **External Sensors (Outside-in tracking):** These systems use external cameras or sensors to track the position and movements of the VR headset and controllers.
- **Inside-out tracking:** Uses cameras or sensors integrated into the VR headset itself to track the user's movement without the need for external devices.
- **Key Features:**
 - Provides precise tracking of the user's head and hand movements.
 - Ensures accurate and smooth interaction within the virtual environment.

3. Controllers

- These are handheld devices used by the user to interact with the virtual world.
- They typically include buttons, triggers, touchpads, and haptic feedback for tactile sensations.
- **Key Features:**
 - Motion sensing: Most controllers have sensors (e.g., accelerometers, gyroscopes) to detect hand movements.
 - Haptic feedback: Vibration to simulate touch sensations within the virtual world.
- Often used to interact with virtual objects, navigate menus, or simulate actions like grabbing or shooting

4. Audio System

- VR often incorporates spatial audio to simulate realistic sound that changes based on the user's head orientation.
- **Key Features:**
 - 3D or binaural audio creates the illusion of sound coming from specific directions or distances in the virtual space.
 - Headphones are commonly used to provide high-quality audio, but some headsets have built-in speakers.

5. Motion Tracking (Optional)

- Sometimes VR setups include additional motion tracking hardware, such as full-body suits or external sensors (e.g., Vive trackers, Kinect).
- **Key Features:**
 - Tracks body movement beyond just the hands and head.
 - Allows for more immersive and physically engaging VR experiences.

6. Software

- VR software includes the VR content, environments, and interactive elements that the user experiences.
- **Key Features:**
 - Games, simulations, or educational experiences specifically designed for VR.
 - Physics engines to simulate realistic interactions with virtual objects.
 - Graphics engines to render 3D environments and objects in real-time.

7. Computing Device (PC, Console, or Standalone Device)

- **PC VR:** Powerful PCs with high-end graphics cards are used for VR experiences (e.g., Oculus Rift, HTC Vive).
- **Console VR:** Some systems (like PlayStation VR) require consoles for operation.
- **Standalone VR:** Devices like the Oculus Quest do not require a PC or console and run on their own internal hardware.
- **Key Features:**
 - High-performance processors and graphics cards to render the immersive virtual world smoothly.
 - Connectivity to VR headsets and controllers.

8. Haptic Feedback (Advanced)

- Some VR systems incorporate wearable devices such as gloves or suits that provide haptic feedback to simulate touch sensations.
- **Key Features:**
 - Haptic gloves or full-body suits can simulate textures, temperature, or pressure.
 - Creates a more immersive experience where the user can feel the virtual environment.

9. Eye Tracking (Optional, Advanced)

- Some high-end VR systems use eye-tracking technology to enhance immersion and performance.
- **Key Features:**
 - Tracks where the user is looking and can adjust the display accordingly (foveated rendering) to reduce processing power in areas that aren't being directly viewed.
 - Can enable gaze-based controls or interactions, making the experience more intuitive.
- These components work together to provide an immersive, interactive, and highly engaging experience in virtual reality

TYPES OF VR

- Virtual Reality (VR) can be categorized based on the level of immersion and interaction provided. The main types of VR include:

1. Non-Immersive Virtual Reality

- **Description:** This is the least immersive form of VR where the user interacts with a virtual environment, but the experience doesn't feel like they are fully inside the virtual world.
- **Key Features:**
 - The user views the virtual world on a flat screen, typically a computer monitor or TV.
 - The interaction may occur through a mouse, keyboard, or gamepad.
- **Examples:**
 - Video games on a regular screen.
 - Virtual tours on a desktop computer.
 - Simulations like CAD or training software used on a computer.

2. Semi-Immersive Virtual Reality

- **Description:** This type of VR offers more engagement than non-immersive VR but still doesn't fully envelop the user in a 360-degree experience. Users may interact with a virtual environment, but they are not fully "inside" it.
- **Key Features:**
 - Users typically interact with the VR experience using motion sensors or controllers.
 - It may be viewed on large screens or projected displays, sometimes with elements like stereoscopic 3D.
- **Examples:**
 - Flight simulators with 3D screens.
 - VR setups that use large projection screens or walls in immersive environments.
 - Arcade-style VR experiences in amusement parks or gaming centers.

3. Fully Immersive Virtual Reality

- **Description:** This is the most advanced and engaging type of VR, where the user feels like they are physically inside the virtual environment. It is characterized by high interaction, realistic visuals, and often includes additional sensory feedback.
- **Key Features:**
 - Use of a **VR headset** to display stereoscopic 3D images and surround sound.
 - **Motion tracking** for head, hand, and sometimes full-body movements to interact with the virtual world.
 - **Haptic feedback** (vibration, force feedback) through controllers, gloves, or suits.
 - Users feel fully immersed and able to explore and manipulate the environment.
- **Examples:**
 - VR games (e.g., Oculus Rift, HTC Vive, PlayStation VR).
 - Virtual simulations for training or education, such as flight training or medical procedures.
 - Full-body experiences using motion capture and wearables (e.g., VR arcades with motion platforms).

4. Augmented Reality (AR)

- **Description:** Although AR isn't strictly VR, it overlaps with virtual experiences. AR overlays digital information onto the real world, enhancing the user's physical environment with virtual elements.
- **Key Features:**
 - **Mixed reality** that blends physical and digital worlds.
 - Uses devices like smartphones, tablets, or AR glasses (e.g., Microsoft HoloLens).
 - Users still see their physical surroundings but with additional virtual objects integrated into their view.
- **Examples:**
 - Mobile games like Pokémon GO.
 - AR navigation systems (e.g., Google Maps AR).
 - Interactive product demonstrations in retail or training environments.

5. 360-Degree Video

- **Description:** This type of VR is based on pre-recorded content, typically filmed with specialized 360-degree cameras. Users can look around the video, but they don't interact with the environment like in fully immersive VR.
- **Key Features:**
 - Offers a panoramic, 360-degree video experience.
 - Requires a VR headset to provide full immersion.
 - Mostly passive: the user can look around, but they can't interact with the environment.
- **Examples:**
 - Virtual tours of real-world locations (e.g., museums, landmarks).
 - VR documentaries or concerts.
 - Real-time event experiences like sports games.

6. Cave Automatic Virtual Environment (CAVE)

- **Description:** A CAVE is a fully immersive VR environment, but instead of using a headset, it uses large, multi-projected screens on the walls, floors, and ceiling, where users wear 3D glasses to experience the environment.
- **Key Features:**
 - Multi-wall projection system for immersive experiences.
 - Users wear special glasses to view 3D images projected on the walls and floor.
 - Interaction with the virtual environment is often achieved via controllers or motion tracking.
- **Examples:**
 - Scientific research, such as in architecture or medical fields.
 - Design and visualization environments for simulations.

7. Desktop VR

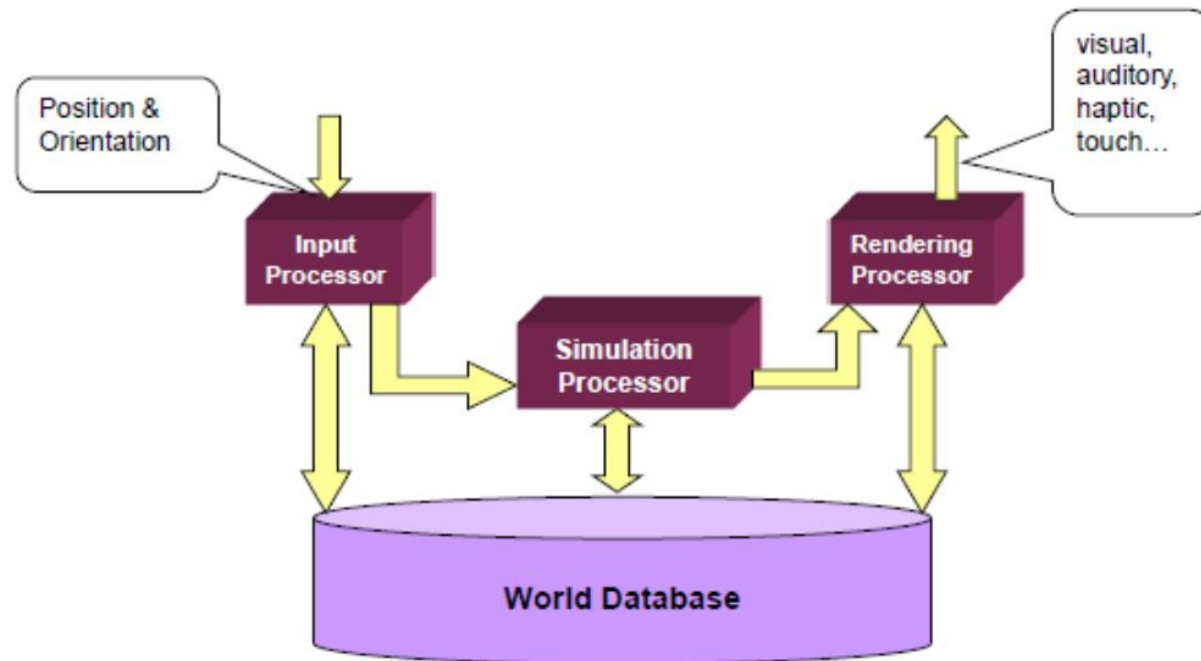
- **Description:** This type uses a standard computer or gaming system to interact with VR-like environments through a screen (non-immersive) or with external accessories like a VR headset or controllers.
- **Key Features:**
 - The user interacts with the virtual world using a mouse, keyboard, or joystick on a computer.
 - Limited immersion compared to fully immersive VR, but allows for interaction in a controlled virtual environment.
- **Examples:**
 - Virtual worlds like Second Life or simulation programs.
 - CAD software for virtual design and modeling.

8. Mobile VR

- **Description:** Mobile VR refers to virtual reality experiences that use smartphones as the primary display and processing unit, typically paired with a VR headset.
- **Key Features:**
 - Inexpensive and portable.
 - Uses the phone's screen and sensors (accelerometer, gyroscope) for display and movement tracking.
 - Less powerful than PC VR but still offers a significant level of immersion.
- **Examples:**
 - Google Cardboard.
- Samsung Gear VR

Architecture of VR System

Input Processor, Simulation Processor, Rendering Processor and World Database



Input Processor

- Control the devices used to input information to the computer. The object is to get the coordinate data to the rest of the system with minimal lag time.
- keyboard, mouse, 3D position trackers, a voice recognition system, etc.

Simulation Processor

- Core of a VR system.
- Takes the user inputs along with any tasks programmed into the world and determine the actions that will take place in the virtual world.

Rendering Processor

- Create the sensations that are output to the user.
- Separate rendering processes are used for visual, auditory, haptic and other sensory systems. Each renderer take a description of the world stat from the simulation process or derive it directly from the World Database for each time step.

World Database (World Description Files)

- Store the objects that inhabit the world, scripts that describe actions of those objects.

3D Position Trackers

- Positional tracking detects the precise position of the head mounted displays, controllers, other objects or body parts within Euclidean space.
- Positional tracking registers the exact position due to recognition of the rotation (pitch, yaw and roll) and recording of the translational movements.
- Since virtual reality is about emulating and altering reality it's important that we can track accurately how objects (like the head or the hands) move in real life in order to represent them inside VR. Defining the position and orientation of a real object in space is determined with the help of special sensors or markers.
- Sensors record the signal from the real object when it moves or is moved and transmit the received information to the computer.
- Wireless tracking • Optical tracking • Tracking with Markers • Marker less Tracking • Inertial Tracking • Sensor Fusion • Acoustic Tracking • Magnetic Tracking

Application of VR

- ☐ Education and training
- ☐ Video Games
- ☐ Fine art
- ☐ Heritage and archaeology
- ☐ Architecture design
- ☐ Urban Design
- ☐ Therapy
- ☐ Theme parks
- ☐ Concerts
- ☐ Retail
- ☐ Film
- ☐ Media
- ☐ Motion picture

Augmented Reality (AR)

- **Augmented Reality (AR)** is a technology that overlays digital information—such as images, videos, and 3D models—onto the real world, enhancing the user's perception and interaction with their environment. Unlike virtual reality (VR), which creates a completely immersive digital experience, AR integrates digital content with the physical world, allowing users to see both simultaneously.



Key Concepts of Augmented Reality:

1. Components of AR:

- ❑ **Hardware:** Devices like smartphones, tablets, and AR glasses (e.g., Microsoft HoloLens, Magic Leap) that enable AR experiences.
- ❑ **Software:** Applications and platforms that create AR content, such as ARKit (iOS), ARCore (Android), and Unity for game development.
- ❑ **Sensors and Cameras:** Used to detect and analyze the environment, allowing digital content to be placed accurately in real-world settings.

2. Types of AR:

- ❑ **Marker-Based AR:** Uses visual markers (like QR codes) to trigger the display of digital content.
- ❑ **Markerless AR:** Utilizes GPS, accelerometers, and compasses to provide a more immersive experience without the need for specific markers (e.g., Pokémon GO).
- ❑ **Projection-Based AR:** Projects digital images onto real-world surfaces.
- ❑ **Superimposition-Based AR:** Replaces a view of an object with an augmented view of it (e.g., overlaying furniture in a room).

3. Applications of AR:

- ❑ **Entertainment:** Video games (like Pokémon GO), interactive storytelling, and immersive theater experiences.
- ❑ **Education:** Interactive learning experiences that enhance textbooks or help with complex subjects (e.g., anatomy simulations).
- ❑ **Retail:** Virtual try-ons for clothing or makeup and visualizing products in a user's space (e.g., IKEA Place app).
- ❑ **Healthcare:** Surgical training and simulations, as well as visualizing anatomy with enhanced details.
- ❑ **Maintenance and Repair:** Providing technicians with real-time information overlaid on machinery.

Mixed Reality (MR)

Mixed Reality (MR) is a blend of the physical and digital worlds, allowing users to interact with both physical and virtual objects in real-time. Unlike Augmented Reality (AR), which overlays digital information on the real world, and Virtual Reality (VR), which creates an entirely immersive digital environment, MR enables a more integrated experience where digital and physical elements coexist and interact.



Key Features of Mixed Reality in Computer Graphics:

- ❑ **Real-Time Interaction:** MR allows users to interact with both real and virtual objects in real-time, creating a more immersive experience.
- ❑ **Spatial Awareness:** MR systems recognize the physical environment and can place digital content in a way that interacts meaningfully with real-world objects. This requires sophisticated depth sensing and spatial mapping technologies.
- ❑ **Environment Mapping:** MR applications often utilize 3D mapping of the physical environment, which helps in accurately positioning and anchoring virtual objects within the real world.
- ❑ **Natural User Interfaces:** MR often employs gesture recognition, voice commands, and eye tracking to enable intuitive interactions, allowing users to manipulate digital content as they would physical objects.

Applications of Mixed Reality in Computer Graphics:

- ❑ **Gaming:** MR games allow players to interact with virtual characters and elements within their physical space, providing a more immersive gaming experience.
- ❑ **Education:** MR can create interactive learning environments where students can visualize complex concepts, such as anatomy or physics, in a more engaging way.
- ❑ **Healthcare:** Surgeons and medical professionals can use MR to overlay critical data on patients or visualize anatomy during procedures.
- ❑ **Design and Prototyping:** Architects and designers can visualize and modify 3D models of buildings and products in real-world contexts, facilitating better decision-making and collaboration.
- ❑ **Remote Collaboration:** MR enables teams to work together on projects, visualizing and manipulating 3D models in a shared space, regardless of their physical locations.

Conclusion

Mixed Reality represents a significant advancement in computer graphics, combining elements of AR and VR to create experiences that are more interactive and immersive. With ongoing developments in hardware and software, MR is poised to transform various sectors, offering new ways to visualize, design, and interact with both digital and physical worlds.

Feature	Augmented Reality (AR)	Virtual Reality (VR)	Mixed Reality (MR)
Definition	Overlays digital content on the real world	Fully immersive digital environment	Blends physical and digital worlds where they interact
User Experience	Awareness of real environment with digital overlays	Complete immersion in a virtual space	Interaction with both real and virtual objects
Technology	Smartphones, tablets, AR glasses	VR headsets, motion tracking	Advanced MR headsets with sensors and cameras
Applications	Mobile games, navigation, retail	Gaming, training simulations	Collaborative design, remote assistance
Immersion Level	Low to moderate	High	Moderate to high

Game Development and Real-Time Graphics

Introduction

- ❑ Game development is the process of designing, programming, and producing video games, while real-time graphics refers to the rendering of visual content instantly to provide smooth, interactive experiences.
- ❑ Modern game development relies on real-time graphics to create immersive environments and responsive gameplay.
- ❑ Game development with real-time graphics is a constantly evolving field, blending technology, creativity, and engineering. The goal is always to immerse players by rendering stunning, responsive environments, characters, and worlds. Real-time graphics make the gaming experience interactive and realistic, pushing boundaries for better visuals and smoother gameplay.

1. Game Development Pipeline

1.1. Pre-Production

- ❑ Concept & Storyboarding: Defining game mechanics, storyline, and characters.
- ❑ Market Research: Identifying the target audience and game genre.
- ❑ Technical Planning: Deciding on game engines, programming languages, and tools.

1.2. Production

- ❑ Game Design: Creating game mechanics, levels, and UI/UX design.
- ❑ Asset Creation: Designing 2D/3D models, textures, and animations.
- ❑ Programming & AI: Implementing gameplay logic, physics, and artificial intelligence.
- ❑ Networking & Multiplayer: Enabling real-time interactions in online games.
- ❑ Sound Design & Music: Composing sound effects and background scores.

1.3. Testing & Debugging

- ❑ Playtesting: Identifying bugs and refining gameplay mechanics.
- ❑ Performance Optimization: Reducing lag, optimizing frame rates, and enhancing efficiency.

1.4. Deployment & Post-Release

- ❑ Publishing: Releasing the game on platforms like PC, console, or mobile.
- ❑ Updates & DLCs: Providing patches, additional content, and bug fixes.

2. Real-Time Graphics in Game Development

- ❑ Real-time graphics enable games to render dynamic scenes instantly.
- ❑ Key components include:

2.1. Rendering Pipeline

The rendering pipeline processes game visuals in three main stages:

- 1.Application Stage:** Processes user input and game logic.
- 2.Geometry Stage:** Converts 3D objects into a 2D representation.
- 3.Rasterization & Shading Stage:** Applies textures, lighting, and effects.

2.2. Graphics Engines

- ❑ Popular game engines with real-time rendering capabilities:
- ❑ Unreal Engine: High-end graphics, photorealistic rendering, and advanced physics.
- ❑ Unity: Flexible engine for 2D/3D games, AR/VR, and mobile games.
- ❑ Godot: Open-source game engine with real-time rendering support.
- ❑ CryEngine: Known for realistic visuals and physics-based rendering.

2.3. Real-Time Rendering Techniques

- ❑ Rasterization: Converts 3D objects into 2D pixels for display.
- ❑ Ray Tracing: Simulates real-world lighting and shadows for realism.
- ❑ Global Illumination: Enhances lighting by simulating light bounce and reflections.
- ❑ Level of Detail (LOD): Reduces rendering complexity for distant objects.
- ❑ Shaders: Custom programs that control pixel rendering, lighting, and materials.